



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
11-07-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

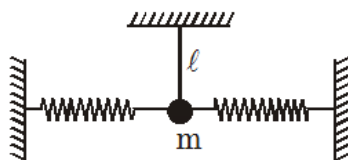
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

PHYSICS

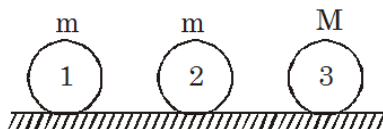
SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

1. Find the angular frequency of small oscillations of the pendulum, shown in figure. The length of thread $\ell = 1$ m, the mass of the ball $m = 1$ kg each spring stiffness $k = 7.5$ N/m. The oscillations occur in the plane of the paper.



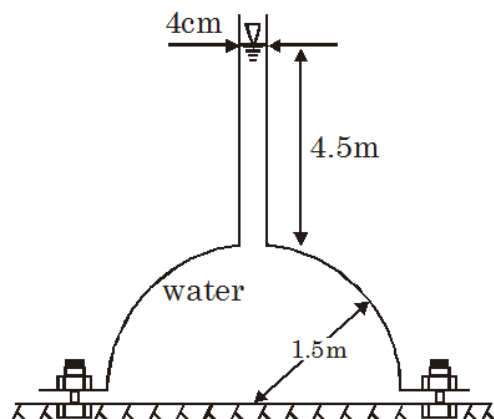
- (A) 5 rad/s (B) 15 rad/s (C) 10 rad/s (D) 7.5 rad/s
2. Two balls marked 1 and 2 of the same mass m and a third ball marked 3 of mass M are arranged over a smooth horizontal surface as shown in the figure. Ball 1 moves with a velocity v towards balls 2. All collisions are assumed to be elastic. If $M < m$ the number of collisions between the balls will be :



- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

3. The half sphere container in the figure is filled with water and fixed to the floor by two bolts (one on each side of the sphere). How large is the force in each bolt that is required to hold the sphere down if its weight is $25 \pi \text{ kN}$?



- (A) $87.5 \pi \text{ kN}$ (B) $43.75 \pi \text{ kN}$ (C) $62.5 \pi \text{ kN}$ (D) $24.35 \pi \text{ kN}$
4. A mercury barometer (which is basically a vertical tube whose one end is open and dipped in mercury, the other end of the tube is closed) whose scale is on the stand behind the glass tube, reads 740 mm. Because of the low reading, it is suspected that some air is present in the space above the mercury. The space is 60 mm long. The open end of the barometer is lowered farther into the mercury reservoir. When the barometer reading is 730 mm, the space above the mercury is 40 mm long. What is the true atmospheric pressure?
- (A) 745 mm of Hg column (B) 750 mm of Hg column
(C) 755 mm of Hg column (D) 760 mm of Hg column

Space for Rough Work

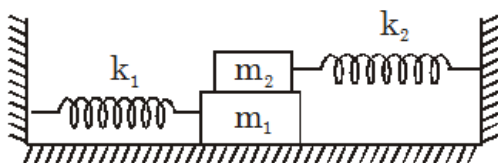
SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. A spring of spring constant 200 N/m has a mass 0.5 kg attached to one end. The other end of the spring is attached to a smooth horizontal table. The mass is pulled so that the spring is extended by an amount 0.05 m. If at this moment ($t = 0$), the mass is projected with a speed of 1 m/s in the direction of increasing extension of spring :
 (A) At $t = \pi/20$ sec, spring has an extension of 5cm.
 (B) The maximum extension of the spring occurs at $\pi/80$ sec.
 (C) The maximum velocity is $\sqrt{2}$ m/s.
 (D) The maximum extension is $5\sqrt{5}$ cm.

Space for Rough Work

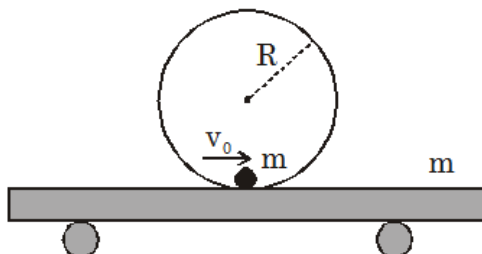
2. A simple pendulum of length ℓ is undergoing oscillations in a vertical plane with a small amplitude and energy E_0 . When the bob was passing through the mean position, suddenly the length of the thread was increased by $\Delta \ell$. When it reaches the extreme position, the length of thread was restored to the original value. $\Delta \ell \ll \ell$.
- (A) At the end of both the steps, the energy of SHM is more than E_0 .
- (B) At the end of both the steps, the energy of SHM is less than E_0 .
- (C) At the end of 1st step, the energy of SHM is more than E_0 .
- (D) At the end of 1st step, the energy of SHM is less than E_0 .
3. Two blocks of mass m_1 and m_2 ($m_1 > m_2$) are performing SHM together with same amplitude and same time period shown. Surface m_1 and ground is smooth, while between m_1 and m_2 the coefficient of friction is C . Given that $\frac{k_1}{m_1} > \frac{k_2}{m_2}$, choose the correct options :



- (A) Time period of SHM is $2\pi\sqrt{\frac{\mu}{k}}$, where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ and $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$.
- (B) Time period of SHM is $2\pi\sqrt{\frac{\mu}{k}}$, where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ and $k = k_1 + k_2$.
- (C) Time period of SHM is $2\pi\sqrt{\frac{\mu}{k}}$, where $\mu = m_1 + m_2$ and $k = k_1 + k_2$.
- (D) Maximum possible amplitude of this SHM is $\frac{2Cm_2g(m_1 + m_2)}{|k_1m_2 - k_2m_1|}$.

Space for Rough Work

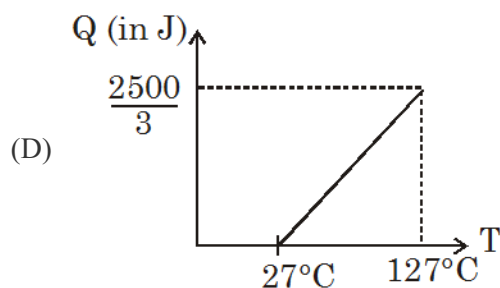
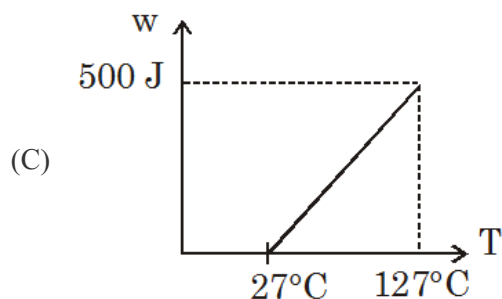
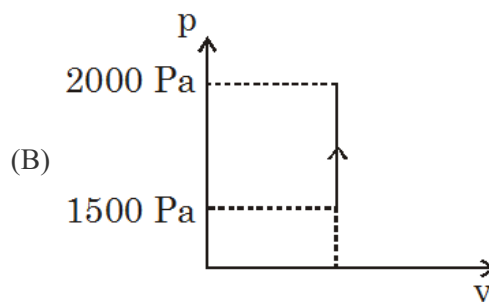
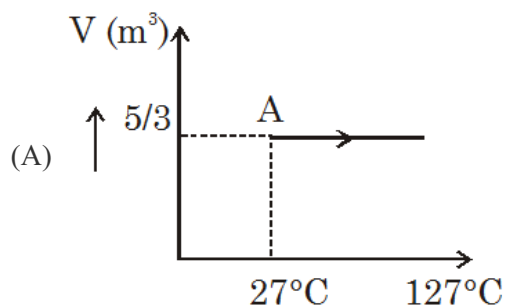
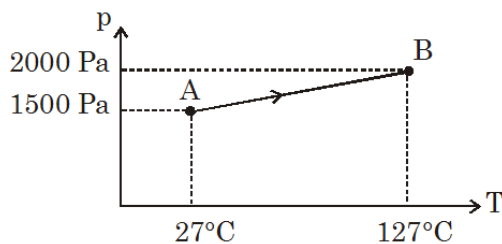
4. The total mass of the trolley, which has small light wheels, and the ring of radius $R = 1\text{ m}$ fixed on the trolley is m (see the figure). A small point-like object of mass m is placed to the bottom fixed of the ring. The small object is given an initial speed of v_0 . The trolley just rises from the ground when the object reaches the topmost point of the ring. Friction is negligible everywhere.



- (A) As the object moves on the ring, the linear momentum of trolley and object is conserved.
 (B) As the object moves on the ring, the mechanical energy of trolley and the object is conserved.
 (C) The value of v_0 is $\sqrt{10gR}$.
 (D) The speed of the trolley goes on increasing till the object reaches the top of the ring.

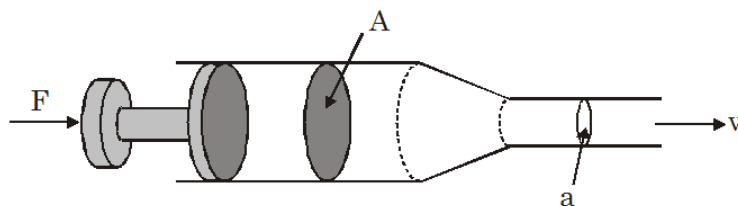
Space for Rough Work

5. One mole of a sample of monoatomic gas is subjected to the process as shown in the diagram below. Choose the correct Pressure-Volume diagram, Volume-temperature diagram, work done by gas-temperature diagram, heat given-temperature diagram.

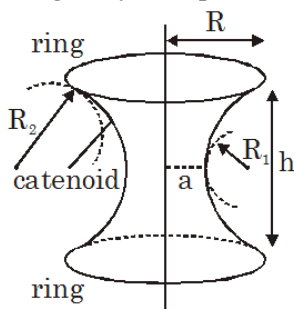


Space for Rough Work

6. In North India people observe a celebrated festival known as Holi. There we spray our friends with colored dyes using a device known as a "Pichkari". Assume that you are holding a horizontal pichkari whose one side is fitted with piston of area A . The other side has a cone fitted in the pichkari. The cone has a narrow opening of area a . The piston is moving with speed v_0 .



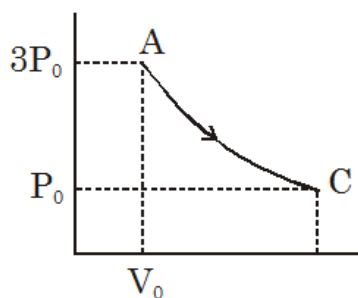
- (A) $v = \frac{Av_0}{a}$ (B) $F \approx \frac{1}{2}\rho Av^2$ (C) $F \approx \frac{1}{2}\rho Av_0^2$ (D) $F \approx \frac{1}{2}\rho av^2$
7. Two rings of radius R are covered by a flat soap film. Now another soap film is made between the two rings whose shape is as shown. This shape is called a catenoid. The catenoid is symmetric about the common axis of the rings. The minimum radius of the catenoid is at the center. This radius $a = R/2$. Soap film has a surface tension S . Neglect gravity. The pressure outside is atmospheric (P_0).



- (A) The pressure inside is also P_0 .
- (B) The Radius of curvature of the catenoid R_1 at the center is R .
- (C) The radius of curvature of the catenoid R_2 at the edge is $2R$.
- (D) The film is shaped like a catenoid because it has the smallest area in comparison to other surfaces under similar conditions.

Space for Rough Work

8. The pressure and volume of a monoatomic ideal gas are related as $P \propto \frac{1}{V^2}$ for a process $A \rightarrow C$ as shown in the figure. The pressure and volume at A is $3P_0$ and V_0 respectively and the pressure at C is P_0 .



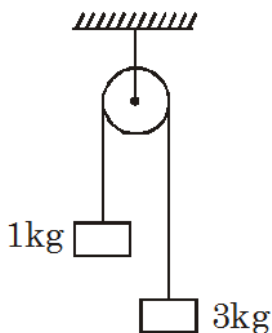
- (A) The work done by the gas in the process $A \rightarrow C$ is $(3 - \sqrt{3})P_0V_0$.
- (B) The molar heat capacity of the ideal gas for the process $A \rightarrow C$ is $R/2$
- (C) Heat is extracted from the gas during the process
- (D) Heat is given to the gas during the process.

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

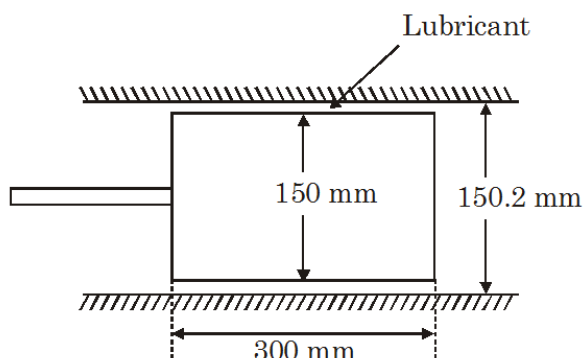
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

- Two particles P_1 and P_2 are performing SHM along the same line about the same mean position. Initially they are at their positive extreme position. If the time period of each particle is 12 sec and the difference of their amplitudes is 12 cm then find the minimum time (in s) after which the separation between the particles become 6 cm.
- In an ideal atwood machine system shown here, find the power (in W) exerted by gravitational force on 3kg block 2 sec after the system is released from rest.

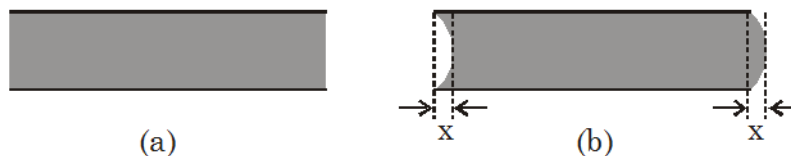


Space for Rough Work

3. A lubricant between a piston and cylinder according to below figure has a kinematic viscosity of $3 \times 10^{-5} \text{ m}^2/\text{s}$ and a relative density of 0.9. If the piston has a mean velocity of 6 m/s what approximate power loss (in kW) will develop due to friction. The piston has a diameter of 150 mm and length 300 mm. The cylinder has a diameter of 150.2 mm.

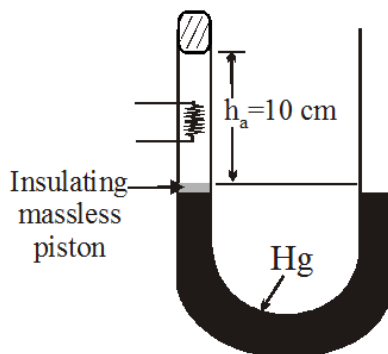


4. A capillary tube of length L and radius r is filled with water as shown in (a). The tube is horizontal. The tube is shaken slightly in the horizontal direction so that it acquires a shape as shown in (b). The radius of curvature of both the surfaces is always equal. If the distance x is very small, find the angular frequency (in rad/sec) of oscillation of the water column. The tube remains stationary while the water column is oscillating. Take surface tension as 0.01 N/m, density of water = 1000 kg/m^3 , $r = 1 \text{ mm}$, $L = 10 \text{ cm}$



Space for Rough Work

5. A U tube, made of heat-insulating material, is shown in figure. Left limb is closed by a non-conducting cork, the temperature of air in left limb is 300 K. How much heat (in Joule) is required to be given by the coil so that the temperature of air rises to 510 K? The thermal expansion of mercury is negligible. (Take : Area of the tube as 0.1 m^2 , $P_{\text{atm}} = 75 \text{ cm of Hg}$, $R = 25/3 \text{ J/mol K}$, specific gravity of mercury = 13.6). Assume that the initial level of mercury in both limbs is same. Air is diatomic gas.



6. A frog in a hemispherical pod finds that he just floats without sinking into a sea having a density of 1.35 g/cm^3 (Fig). If the pod has a radius of 6.00 cm and a negligible mass, what is the mass of the frog (in gm)?



Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
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- Answer to each question will be evaluated according to the following marking scheme :

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. Which of the following 'ic' acid have σ/π ratio > 3 ?
(A) $\text{H}_2\text{S}_2\text{O}_5$ (B) $\text{H}_2\text{S}_2\text{O}_6$ (C) $\text{H}_4\text{P}_2\text{O}_6$ (D) $\text{H}_2\text{S}_2\text{O}_3$
 2. A proton accelerated from rest through a potential difference of 'V' volts has a wavelength λ associated with it. An alpha particle in order to have the same wavelength must be accelerated from rest through a potential difference of -
(A) V volt (B) 4V volt (C) 2V volt (D) $\frac{V}{8}$ volt
 3. 0.7 g of an organic compound on heating evolved all of its nitrogen into ammonia which is reacted with 200 ml of 0.25 M HNO_3 solution. The residual HNO_3 solution is reacted completely with 30 ml of 0.5 M Ca(OH)_2 . % by mass of nitrogen in the sample is :
(A) 20 (B) 15 (C) 30 (D) 40
 4. Select the most basic compound in aqueous medium out of following :
(A) NH_3 (B) MeNH_2 (C) Me_2NH (D) Me_3N
-

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

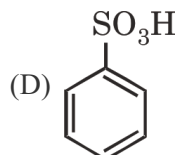
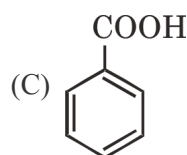
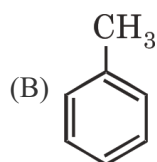
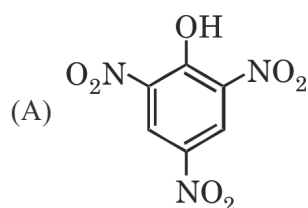
1. Select the correct order against the stated property :-
 (A) Glycerol > Ethylene glycol : Viscosity
 (B) Ion-dipole > Dipole-dipole : Exponent of 'r' in their energy distance function
 (C) $x > y$: $(\text{Si}_x\text{O}_{11})^{6-}$, $(\text{Si}_y\text{O}_{15})^{10-}$
 (D) $\text{CH}_2\text{Cl}_2 > \text{CHCl}_3$: Dipole moment

Space for Rough Work

2. 4 % s-character in P-H bond in PH_3 indicates that :-
- (A) Lone pair resides in stereo chemically inactive 3d-orbital as P belongs to 3rd period
- (B) P-H bond length is longer in this case as compared to, if hybridisation would have taken place
- (C) Bond angle in this case is less as compared to, if hybridisation would have taken place
- (D) The molecule will not be polar
3. Which of the following correct order against the stated property is due to Vander Waal forces of attraction?
- (A) $\text{I}_3^- > \text{Br}_3^- > \text{Cl}_3^-$: order of stability
- (B) $\text{NH}_4\text{Cl} > \text{NaCl}$: Solubility in polar solvent
- (C) $\text{H}_2(\ell) > \text{He}(\ell)$: BP
- (D) $\text{SiO}_2 > \text{NaCl}$: MP
4. 90 gm glucose is dissolved in 800 gm water to get a solution of density $\frac{89}{80}$ gm/ml. The correct concentration of solution is /are -
- (A) $\frac{90}{8}\%$ (w/w) (B) $\frac{90}{8}\%$ (w/v) (C) 0.625 m (D) 0.625 M

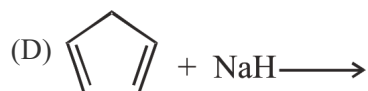
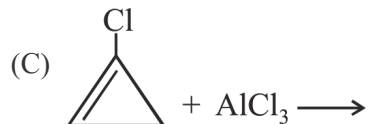
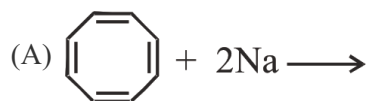
Space for Rough Work

5. Light from a discharge tube containing hydrogen atom fall on the surface of a piece of sodium. The kinetic energy of a fastest photo electron emitted from sodium is 0.73 eV & work function for sodium is 1.82 eV (collector plate is connected to +ve terminal of battery having voltage 2 volts.), which of the following are true regarding above experiment.
- (A) The energy of photon causing the photoelectric effect is 2.55 eV
- (B) Hydrogen atoms electron undergoes a transition from $n = 4$ to $n = 2$ to show given photoelectric effect.
- (C) Change in angular momentum of the electron in the hydrogen atom will be $\frac{h}{\pi}$
- (D) Kinetic energy of photoelectron at emitter plate will be 0.73 eV & at collector plate will be 2.73 eV
6. Compounds which are more acidic than phenol are :



Space for Rough Work

7. Which of the following reaction will undergo in forward direction ?



8. Which of the following IUPAC name is/are incorrect.

(A) 1-Amino pentan-1-one

(B) 2-Methyl pentan-5-ol

(C) 3-Carboxy pentane

(D) 1-Bromo-3-chloro-4-iodo cyclohexane

Space for Rough Work

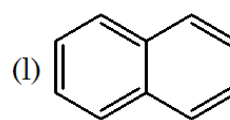
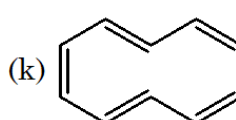
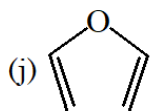
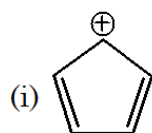
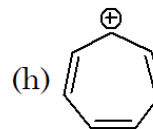
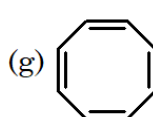
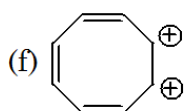
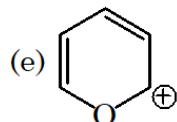
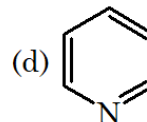
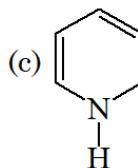
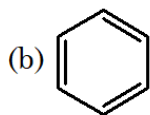
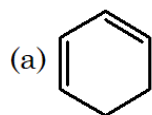
SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. Numbers of salts having more than 2 ions among the following :-
 Potassium hypobromite, Barium hypophosphite, Sodium hydrogen pyrophosphate, Sodium trimetaborate, Calcium dihydrogen orthophosphate, Sodium Trimetaphosphate
2.
$$\frac{\text{Number of six membered rings} - \text{number of five membered rings}}{\text{number of hybrid orbitals of each carbon}}$$
 in fullerene (C_{60}) = ?
3. 1 mole each of A & D is introduced in 1 ℓ container simultaneous two equilibria are established:
 $A \rightleftharpoons B + C \quad K_C = 10^6 M^2$
 $B + D \rightleftharpoons A \quad K_C = 10^{-6} M^{-1}$
 If equilibrium concentration of A is approximately 10^{-x} then what will be the value of x (x is an integer).
4. Total number of structural isomers of $C_4H_{11}N$.

Space for Rough Work

5. How many of the following are aromatic :-



6. Density of a Vander Waal's gas at 500 K and 1 atm is found to be 0.8 kg/m^3 . It is also found that the gas diffuses $\sqrt{\frac{3}{2}}$ times slower than pure O_2 gas under identical conditions. The value of Z for the real gas is ($R = 0.08 \text{ atm-l/mol-K}$) :

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Number of solutions of $\cos^4 2x + 2\sin^2 2x = 17(\cos x + \sin x)^8$ is, $0 < x < 2\pi$
 (A) 2 (B) 4 (C) 6 (D) 3
2. Tangents from point $P(8\sqrt{2}, 6\sqrt{2})$ to the circle $C_1: x^2 + y^2 - 8\sqrt{2}x - 6\sqrt{2}y + 25 = 0$, meet at point A & B and go on to intersect the tangents from the point $R(0, 0)$ to circle C_1 at points Q and S respectively. Tangents SR and QR meet C_1 at C and D respectively. The points A, B, C, D in order are all joined to each other to form a quadrilateral ABCD with intersecting diagonals (where E is the point of intersection of AC & BD) giving rise to four triangles EAB, EBC, ECD, EDA. Find the area of the quadrilateral formed by joining the incentres of the triangles EAB, EBC, ECD, EDA.
 (A) $50(3 + 2\sqrt{2})$
 (B) $50(3 - 2\sqrt{2})$
 (C) $50(9 - 4\sqrt{2})$
 (D) $(9 + 4\sqrt{2})$

Space for Rough Work

3. Let $S = \sum_{r=1}^{16} \left[\frac{(1 + \tan 4\theta \cdot \tan \theta)}{r} \sum_{k=1}^{3r} \tan\left(\theta + \frac{k\pi}{3}\right) \right]$ where $\theta = 4^\circ$. Then S is equal to
- (A) $\tan 4^{16} - \tan 4$
 (B) $3(\tan 4^{16} - \tan 4)$
 (C) $\tan 4^{17} - \tan 4$
 (D) $3(\tan 4^{17} - \tan 4)$
4. Let a tangent to parabola $y^2 = 4ax$ at point P (other than vertex), intersects its directrix at point Q and S represent focus of parabola. If normal to parabola at point P intersects line joining Q and S in R then the value of $\frac{(PS)(QR)}{(PQ)(PR)}$.
- (A) 1
 (B) 2
 (C) $\frac{3}{2}$
 (D) $\frac{1}{2}$

Space for Rough Work

SEC-2 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

- The equation of tangent at P(4, 3) on the parabola whose axis is x-axis is $2y - x - 2 = 0$, then which of the following is/are true?
 (A) Equation of focal chord is $4x - 3y - 7 = 0$
 (B) Focus is $(\frac{7}{4}, 0)$
 (C) Directrix is $x = \frac{1}{4}$
 (D) None of these

Space for Rough Work

2. Which of the following is true regarding the solution set of equation

$$\frac{\log_x (2a - x)}{\log_x 2} + \frac{\log_a x}{\log_a 2} = \frac{1}{\log_{(a^2-1)} 2}$$

- (A) $a \leq 1 \Rightarrow x \in \phi$
 (B) $a > 1, a \neq \sqrt{2}, 2 \Rightarrow x = a \pm 1$
 (C) $a = 2 \Rightarrow x = 3$
 (D) $a > 1, a \neq \sqrt{2}, 2 \Rightarrow x = a - 1$
3. Focus divide focal chord PQ of parabola in the ratio $K_1 : K_2$. The perpendicular distance of P from directrix is ℓ_1 and tangent at Q cuts directrix at R such that $RQ = \ell_2$. Then which of the following is/are true.

- (A) Distance of focus from tangent at P is $\frac{K_1 \ell_2}{K_1 + K_2}$
 (B) Distance of focus from tangent at Q is $\frac{(K_1 + K_2)}{4K_2}$
 (C) Length of latus rectum is $\frac{4K_2 \ell_1}{K_1 + K_2}$
 (D) Length of latus rectum is $\frac{4K_1 \ell_2}{K_1 + K_2}$

Space for Rough Work

4. Let A and B be variable points on the parabola P, such that the tangent at A and B are perpendicular to each other. The locus of the centroid of the triangle formed by A, B and the vertex of P is the parabola P_1 . Apply the same process to P_1 , to obtain the parabola P_2 and repeat the process to obtain all together the sequence of parabolas $P, P_1, P_2, \dots, P_n$. If the equation of P is $y^2 = ax$ then which of the following is/are true.
- (A) Focus of P_7 is $(\frac{a}{4}, 0)$
- (B) Sum of lengths of Latus Rectum of Parabolas $P_1, P_2, \dots, P_n$ is $\frac{a}{2}(1 - \frac{1}{3^n})$
- (C) Sum of lengths of Latus Rectum of Parabolas $P_1, P_2, \dots, P_n$ is $a(1 - \frac{1}{3^n})$
- (D) Vertex of parabolas P_5 is $(\frac{a}{2}(1 - \frac{1}{3^5}), 0)$
5. Let $\{a_n\}$ a G.P. such that arithmetic mean of first three terms is equal to $\frac{7}{6}$ of their geometric mean and the first term of the GP is a natural number and $[\sqrt{\sqrt{a_2}}] = 1$. Identify which of the following statement(s) is/are correct ? [Note : $[k]$ denotes greatest integer less than or equal to k.
- (A) possible number of values of a is 6^2
- (B) sum of all possible common ratios is $\frac{5}{2}$
- (C) a_3 can not take value 3
- (D) a_3 can not take value 9

Space for Rough Work

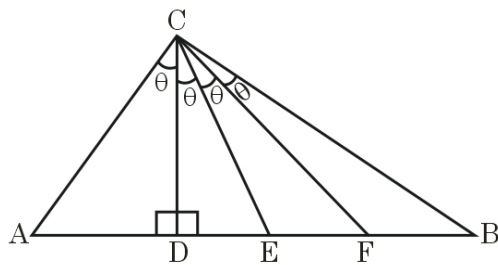
6. The line $2x - y = 1$ intersect the parabola $y^2 = 4x$ at the points A and B and the normals at A and B intersect each other at the point G. If a third normal to the parabola through G meets the parabola at C then which of the following statement(s) is/are correct.
- (A) sum of the abscissa and ordinate of the point C is -1 .
- (B) the normal at C passes through the lower end of the latus rectum of the parabola.
- (C) centroid of the triangle ABC lies at the focus of the parabola.
- (D) normal at C has the gradient -1 .

7. If $D_1 = \begin{vmatrix} -(2^8 + 1)^2 & (2^{16} - 1) & \frac{1}{2^8 - 1} \\ 2^{16} - 1 & -(2^8 - 1)^2 & \frac{1}{2^8 + 1} \\ \frac{1}{2^8 - 1} & \frac{1}{2^8 + 1} & \frac{1}{(2^{16} - 1)^2} \end{vmatrix}$ and $D_2 = \begin{vmatrix} -(3^{12} + 1)^2 & 3^{24} - 1 & \frac{1}{3^{12} - 1} \\ 3^{24} - 1 & -(3^{12} - 1)^2 & \frac{1}{3^{12} + 1} \\ \frac{1}{3^{12} - 1} & \frac{1}{3^{12} + 1} & \frac{1}{(3^{24} - 1)^2} \end{vmatrix}$,
then

- (A) $D_1 = 2$ (B) $D_2 = 9$ (C) $D_1 = 4$ (D) $D_1 = D_2$

Space for Rough Work

8. If in a triangle ABC, the altitude CD, angle bisector CE and median CF from vertex C divides $\angle ACB$ in four equal angles as shown in figure then what can be said about $\triangle ABC$?



- (A) It is acute angled triangle
- (B) It is obtuse angled triangle
- (C) Incentre, centroid and orthocentre, circumcentre lies inside the triangle
- (D) It is right angled triangle

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

-
1. The sum of all solutions of the equation $\frac{1}{x^2-1} + \frac{2}{x^2-2} + \frac{3}{x^2-3} + \frac{4}{x^2-4} = 2010x - 4$ is $\frac{n}{1005}$. Find the value of $\frac{n}{4}$.
 2. $P(3 + 2t, 0)$ and $Q(4, 2 - 1/t)$ where $t < 0$ are two points on a parabola having focus $S(3, 2)$ and vertex at a distance of 5 unit from S . Find the least value of PQ .
 3. ΔABC is a triangle in which $\sqrt{3}\sin C = 2\sec A - \tan A$ and one of side is of length 2. If the maximum area of triangle is λ then find $\frac{[\lambda]}{4}$. (where $[.]$ denotes greatest integer function)
-

Space for Rough Work

4. A variable line is drawn through the origin O. Two points A and B are taken on the line such that $OA = 1$ units and $OB = 2$ units and A and B are on the same side of origin. Through the points A and B two lines are drawn making equal angle α with line AB. If the locus of point of intersection of lines is $x^2 + y^2 = \frac{p + \tan^2 \alpha}{q}$ then determine $\frac{(p - q)}{4}$.
5. The sum of $\frac{1}{2\sqrt{1} + 1\sqrt{2}} + \frac{1}{3\sqrt{2} + 2\sqrt{3}} + \frac{1}{4\sqrt{3} + 3\sqrt{4}} + \dots + \frac{1}{25\sqrt{24} + 24\sqrt{25}}$ is
6. If the circle C_1 touches x-axis and the line $y = x \tan \theta$. $\theta \in (0, \frac{\pi}{2})$ in first quadrant and circle C_2 touches the line $y = x \tan \theta$, y-axis and circle C_1 in such a way that ratio of radius of C_1 to radius of C_2 is 2 : 1, then value of $\tan \frac{\theta}{2} = \frac{\sqrt{a} - b}{c}$ where a, b, c are relatively prime natural numbers. Find $\frac{a + b + c}{4}$.

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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